

Cyclic Groups (Section 6)

Def: A group G is **cyclic** if $\exists a \in G$ such that $\langle a \rangle = G$.

$\{ \overset{e}{a^0}, a^1, a^2, \dots \}$

Thm: Every cyclic group is abelian.

Pf: Let G be cyclic and let $a \in G$ be a generator, so $\langle a \rangle = G$.
Let $b, c \in G$. We know $b = a^n$ and $c = a^m$ for some $m, n \in \mathbb{Z}$.

Then

$$b * c = a^n * a^m = a^{n+m} = a^{m+n} = a^m * a^n = c * b. \quad \square$$

Division algorithm for $\mathbb{Z}$

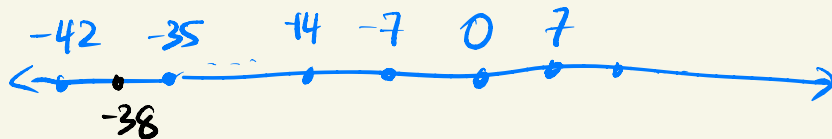
there exist unique

Thm: If $n \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$, then $\exists!$ $q, r \in \mathbb{Z}$ such that

$$n = mq + r \quad \& \quad 0 \leq r < m$$

quotient *remainder*

Ex: $n = -38$, $m = 7$



$$-38 = -42 + 4$$

$$q = -6, \quad r = 4$$

Thm: Any subgroup of a cyclic group is cyclic.

Pf: $\S$ G is cyclic and $\langle a \rangle = G$ and $H \leq G$.

If $H = \{e\}$ then $H = \langle e \rangle$ is cyclic. Otherwise H contains some non-identity element, so let m be the smallest element of $\mathbb{Z}^+$ such that $a^m \in H$. We want to show $H = \langle a^m \rangle$. Let $b \in H$. Write $b = a^n$ for some n . Use the division algorithm to uniquely write

$$n = qm + r, \text{ where } 0 \leq r < m.$$

$$\underbrace{a^n}_{\in H} = \underbrace{(a^m)^q}_{\in H} * a^r, \text{ thus}$$

$$\underbrace{a^{-mq} * a^n}_{\in H} = a^r$$

so by minimality of m , we have $r = 0$. $\square$

Cor: The subgroups of $\langle \mathbb{Z}, + \rangle$ are precisely the groups $\langle n\mathbb{Z}, + \rangle$ for $n \in \mathbb{Z}_{\geq 0}$.

$\uparrow \{c \cdot n \mid c \in \mathbb{Z}\}$

Pf: $\langle \mathbb{Z}, + \rangle$ is cyclic since $\mathbb{Z} = \langle 1 \rangle$. So any $H \leq \mathbb{Z}$ is cyclic $H = \langle n \rangle$ for some n .

Greatest common divisors (gcd's)

Let $r, s \in \mathbb{Z}^+$. Then $H = \{nr + ms \mid n, m \in \mathbb{Z}\}$ is a subgroup of $\langle \mathbb{Z}, + \rangle$. Hence $H = d\mathbb{Z}$ for some $d \in \mathbb{Z}^+$. We call d the **gcd** of r, s and write $d = \gcd(r, s)$.

① closed. $(n_1r + m_1s) + (n_2r + m_2s) = (n_1 + n_2)r + (m_1 + m_2)s \in H$.
② $0 \in H$.
③ $\uparrow$ greatest common divisor $= (nr + ms) = -nr + (m)s \in H$.

Prop: d is a divisor of both r & s and any other such divisor d' must divide d .

Pf: First show $d \mid r$ and $d \mid s$. We know $r \in H = d\mathbb{Z}$ so $d \mid r$. ✓
"divides"
same for s .

Let d' divide both r and s . Since $d \in H$ we can write $d = nr + ms = n(a_1 \cdot d') + m(a_2 \cdot d') = (na_1 + ma_2)d'$ so $d' \mid d$.

Example: $\gcd(42, 72)$

$$42 = 2 \cdot 3 \cdot 7$$

$$72 = 2^3 \cdot 3^2$$

$$d = 2 \cdot 3 = 6$$

Def: Integers r, s are relatively prime if $\gcd(r, s) = 1$.

Example: $r = 12, s = 25$ relatively prime
 $r = 2^2 \cdot 3, s = 5^2$

Example: If $r < p$ & p is prime then r, p are relatively prime.

Prop: If r, s are relatively prime and if $r \mid sm$, then $r \mid m$.
↖ divides

Pf: Book.

We can now describe all cyclic groups.

Thm: Let $G = \langle a \rangle$ be a cyclic group. Then

① If G is infinite, then $G \cong \langle \mathbb{Z}, + \rangle$

② If G is finite, of order n , then $G \cong \langle \mathbb{Z}_n, + \rangle$.

Pf: ① $\S$ G is infinite. Then $a^i \neq a^j$ for $i \neq j$. (for if we did have $a^i = a^j$ with $i < j$, then there are only finitely many elements $a^0, a^1, \dots, a^{j-1}$ in $\langle a \rangle$). So we have a function $f: G \rightarrow \mathbb{Z}$ defined by sending $a^i \mapsto i$, $\forall i \in \mathbb{Z}$.

injective: if $f(a^n) = f(a^m)$, then $n = m$ so $a^n = a^m$

surjective: for any $n \in \mathbb{Z}$, we have $f(a^n) = n$.

homomorphism property:

$$f(a^i * a^j) \stackrel{? \checkmark}{=} f(a^i) + f(a^j)$$

$$\begin{matrix} \text{"} & & \text{"} \\ f(a^{i+j}) & = & i+j \end{matrix}$$

② § G is finite and of order n . $G = \{e, a, a^2, \dots, a^{n-1}\}$
and $a^n = e$.

... (exercise)

Thm: Let $G = \langle a \rangle$ be a cyclic group of order n . Then the subgroup $\langle a^s \rangle$ has order $n/\gcd(s, n)$. Furthermore,

$$\langle a^s \rangle = \langle a^t \rangle \text{ iff } \gcd(s, n) = \gcd(t, n).$$

↑ if and only if

Pf: Book.

Ex: $\mathbb{Z}_4 = \langle 1 \rangle$

$$\langle 0 \rangle = \{0\}$$

$$\langle 1 \rangle = \{0, 1, 2, 3\}$$

$$\langle 2 \rangle = \{0, 2\}$$

$$\langle 3 \rangle = \{0, 3, 2, 1\}$$

$$|\langle 1 \rangle| = 4 = \frac{4}{\gcd(1, 4)}$$

$$|\langle 2 \rangle| = \frac{4}{\gcd(2, 4)} = \frac{4}{2} = 2$$

$$|\langle 3 \rangle| = \frac{4}{\gcd(3, 4)} = \frac{4}{1} = 4$$

$$|\langle 0 \rangle| = \frac{4}{\gcd(0, 4)} = \frac{4}{4} = 1.$$

Cor: If $G = \langle a \rangle$ is a cyclic group of order n , then the elements which generate G are a^s where $s \notin n$ are relatively prime.

Pf: $|\langle a^s \rangle| = \frac{n}{\gcd(s, n)}$ equals $n \iff \gcd(s, n) = 1$.

Example: If n is prime, what are the generators of $\mathbb{Z}_n$?
 m generates $\mathbb{Z}_n$ for all $1 \leq m < n$.

Example: Subgroup diagram of $\mathbb{Z}_{18}$, $18 = 2 \cdot 3^2$

$\gcd(m, 18) = 2$ for $m = 2, 4, 8, 10, 14, 16 \rightsquigarrow$ these elements each generate the same subgroup

subgroup

$$H \leq \mathbb{Z}_{18}$$

with $H \cong \mathbb{Z}_9$.

